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Estimating the size-selection curves of towed gears, traps, nets and hooks

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Abstract

A general statistical methodology for analysis of selectivity data from any type of fishing gear is presented. This formal statistical modelling of selectivity is built on explicit definitions of the selection process and specification of underlying assumptions and limitations, and this gives the resulting estimates of gear selectivity (and possibly fishing power) a clear interpretation. Application of the methodology to studies using subsampled catch data and to towed gears having windows or grids is outlined, and examples applied to passive nets and towed gears are presented. The analysis of data from replicate deployments is covered in detail, with particular regard to modelling the fixed and random effects of between-haul variation. Recent developments on the design of selectivity experiments are introduced and demonstrated.

Key words: experimental design, fishing gear, maximum likelihood, SELECT, size selectivity

Introduction

Fulton (1893) noted an absence of knowledge about the size selectivity of trawl gear, reporting that "The question of the size of mesh of trawl-nets was raised; and the evidence on this point was somewhat conflicting; the general opinion among the practical men being that the size of the mesh of the net had little or no influence on the size of the fish captured, owing to the meshes drawn together by the strain on the net while fishing." This motivated Fulton (1893) to conduct experiments with trawl meshes of four different sizes and with a small-mesh cover attached behind the aft part of the trawl to retain escapees. Fulton's (1893) results established a strong effect of mesh size, evidenced by the proportion of fish retained in the trawl dropping from 33% in a 1.5 inch (knot to knot; 3.8 cm) mesh to just 6% in a 3.0 inch (7.6 cm) mesh.

A century later, fisheries managers need to know more than just the percentage retention of fishing gears. Percentage retention is of little use by itself because it varies with the length distribution of the fished population. For example, percentage retention of a trawl will be lower if juveniles are fished. Also, two gears with the same percentage retention could select fish in very different ways; one might release some fish of all sizes while the other releases all fish below a certain size. In fisheries management, knowledge of size selectivity can be used in:

- estimation of incidental mortality (i.e. mortality of discards and escapees);
- yield-per-recruit analysis;
- age- and length-based population models;
- estimation of population length frequencies, length at age, etc.

Here we review methodology for estimating size-selection curves using data from comparative selectivity studies, that is, catch data obtained by fishing with a modified gear that includes the experimental gear and some variants of it. Inference about selection is made by comparing the size distribution of the catch from each gear variant. The limitation of this experimental methodology is clear – it can only quantify selectivity occurring after contact with the modified gear. Consideration of the entire process of selectivity permits precise definitions of 'size selection' and allows the extent of this limitation (with regard to the management activities listed above) to be examined ('Background', below). (It is an unfortunate fact that failure to formally define size selection has contributed to confusion and unnecessary division in much of the selectivity literature.)

Historically, methods for estimating the size selectivity of fishing gears using comparative data have largely followed one path for towed gears (e.g. trawls and seines) and traps/pots (Pope et al., 1975), and several other paths for gill nets and hooks (Hamley, 1975). (Unless otherwise stated, the terminology 'gill net' is used generically herein to include entangling nets and all other such passive nets.) A new methodology (Millar, 1992; Millar and Walsh, 1992; Wileman et al., 1996) has now been endorsed by ICES (International Council for the Exploration of the Sea) for towed gear (and trap) selectivity studies, and a European Union sponsored research initiative (DG XIV/C1 Study No. 94/118 'Methods and standards for gillnet selectivity research') has recently recommended a methodology for the analysis of gillnet (and hook) selectivity data. This review consolidates these two new methodologies and emphasizes that they are identical, being derived from the same general statistical model for selectivity ('General model',

below). This general selectivity model is also used in the science of resource selection (Manly et al., 1993).

Special cases of the general statistical model were used in gillnet studies by Kirkwood and Walker (1986) and Wulff (1986). However, they received little subsequent attention in the literature, due perhaps to the bewildering collection of alternative methods that have been proposed recently (Rudstam et al., 1984; Boy and Crivelli, 1988; Ralston, 1990; Helser et al., 1991, 1994; Henderson and Wong, 1991; Salvanes, 1991; Borgstrøm and Plahte, 1992; Spangler and Collins, 1992; Kenchington, 1993).

Applied to towed gear studies, the general model is known as SELECT, an acronym derived from Share Each LEngthclass's Catch Total. The 'sharing of catch totals' is an implementation of the general model that has the advantage of greatly reducing the number of parameters to be estimated. It has recently been shown (Millar and Holst, 1997) that the computational complexity of applying the general model to net and hook data can also be reduced by using the SELECT approach, or (often) may be avoided entirely through the use of the built-in capabilities of existing statistical packages ('Fitting the general model', below).

The sound statistical foundation of the general model allows it to formally address existing questions in gear selectivity. These include the incorporation of subsampling from the catch ('General model'), the choice of suitable selection curves and inference regarding fishing powers of gear variants ('Inference', below), the estimation and/or incorporation of between-haul variation ('Variation between replicate deployments', below), and the design of selectivity experiments ('Design of experiments', below).

Background

Selectivity experiments

A selectivity experiment is called **direct** if the size distribution of the population (or subpopulation of marked individuals) being fished is known, because size selection can then be calculated by direct comparison between the size distributions of the population and the catch. Unfortunately, such knowledge can be difficult to obtain and the literature contains only a few true direct studies. These include a gillnet study by Hamley and Regier (1973) that fished a marked population of walleye (*Stizostedion vitreum vitreum*, Percidae), a combined analysis of 137 tagging experiments on Atlantic cod (*Gadus morhua*, Gadidae) spanning a period of 38 years and six gear types (Myers and Hoenig, 1997), and laboratory studies that monitored the escape of crustaceans from pots (Krouse and Thomas, 1975; Krouse, 1978; Everson et al., 1992). Pseudo-direct studies use estimates of the population size distribution obtained by methods other than comparative fishing; for example, from mark–recapture (Borgstrøm, 1989, 1992; Borgstrøm and Plahte, 1992) or acoustic survey (Winters and Wheeler, 1990).

In practice, almost all selectivity studies are comparative (indirect). The usual procedure is to simultaneously fish different variants (in size and/or shape) of the same gear type, usually with equal effort. For example, a longline can be rigged so that the hooks alternate in type. Gill nets can be constructed of several panels, all of the same size and each of a different type mesh, and the order of the panels can be changed on each day of fishing to reduce the effect of any possible preference by the fish for a particular area of the net.

When fishing with towed gears or traps, an experimental gear can be fished alongside a variant of the gear that is designed to prohibit (as much as is possible) the escape of fish of all sizes. Examples of such variants include a towed gear with a small-mesh cod end, or a pot with no escape gaps. These are often referred to as control gears because, in principle, they are nonselective and the size distribution of the catch should therefore be representive of the fish being encountered. Fishing an experimental cod end in conjunction with a control cod-end can be implemented in several ways. For example, the two cod ends can be fished alternately by the same boat, or fished simultaneously using two boats. Trawls can also be modified so that they have two cod ends. These are known as trouser trawls and frequently have a vertical separator panel that separates the flow of fish at the trawl mouth, hopefully before fish can detect any difference between the two cod ends. Each approach has its advantages and disadvantages. The trouser trawl is a major modification to a standard trawl and may not fish as a standard trawl would. The alternate haul and twin gear methods use an unmodified gear, but risk trawling through different populations of fish.

With towed gears, a second type of indirect experiment is to fish an experimental gear with a small-mesh cover over the cod end. With this method, the small-mesh cover captures all fish that pass through the

experimental cod-end mesh, so all fish contacting the gear are observed. However, the cover can affect gear characteristics and fish escape behaviour (Pope et al., 1975, p. 9). Stewart and Robertson (1985) used underwater camera to monitor covered cod-end trawls and suggested methods of construction to minimize the effect of the cover.

Definitions of selection curve

Selection curves quantify the (relative) capture probability of fish of a given size. Intuitively, selection should be expressed as a function of fish girth for meshed gears, and gape of mouth for hooks. However, length is considerably easier to measure, is closely related to both girth and gape, and is commonly used in fisheries models. Selection is therefore most often expressed as a function of length, and this is assumed here, notwithstanding that this work also applies to crustaceans and molluscs, whence it may be length (or width) of carapace/shell that is used.

In its fullest generality, size selection encapsulates all the processes that cause the probability of capture to **vary** with fish size. We implicitly partition the selection process by giving the following three definitions of selection curve, each differing in the population being selected from.

Definition 1: The **population-selection curve**, $s(l)$, is the (relative) probability that a fish of length l from the population is captured.

Definition 2: The **available-selection curve**, $a(l)$, is the (relative) probability that a fish of length l is captured given that it was available to (but possibly avoided) the gear.

Definition 3: The **contact-selection curve**, $r(l)$, is the (relative) probability that a fish of length l is captured given that it contacted the gear.

The contact-selection curve is just a retention curve, but defining it as a selection curve emphasizes that it quantifies the difference in length distribution between the catch and the population of fish coming in contact with the gear. Similarly, the available-selection curve quantifies the differences between the catch and the population of fish available to the gear, and the population-selection curve is relative to the entire population.

The contact-selection curve $r(l)$ is used in fisheries management for assessing the effect of mesh changes on yield (Bennett, 1984; Kuikka et al., 1996), estimating discard levels, and estimating the incidental mortality of discards and of fish escaping the gear (Suuronen, 1995; Chopin et al., 1996).

The population-selection curve $s(l)$ is required for yield-per-recruit and some age- and length-based population models (Myers and Cadigan, 1995), where it represents the exploitation pattern or partial recruitment to the fishery (Doubleday, 1976). Knowledge of $s(l)$ also enables the population length frequency to be estimated by correcting the observed catch length frequency, and this is often the motivation behind gillnet selectivity studies (Spangler and Collins, 1992).

Statistical models used to describe towed gear and trap/pot selectivity typically model the contact-selection curve $r(l)$. In the gillnet and hook selectivity literature, any of the above selection curves can be modelled, and it is often not clear which. In addition, some gillnet studies have treated length distribution curves (of the catch in each mesh size) as though they were selectivity curves (Regier and Robson, 1966; Hamley, 1975; Helser et al., 1991, 1994). This inconsistent usage has contributed unnecessarily to the difference in opinions on the shape of gillnet and hook selection curves.

The three selection curves defined above are related by:

$$a(l) \propto r(l) \times \text{prob(fish of length } l \text{ contacts the gear given that it is available to the gear)} \quad (1)$$

$$s(l) \propto a(l) \times \text{prob(fish of length } l \text{ is available to the gear).} \quad (2)$$

The probabilities on the right-hand side of Expressions 1 and 2 depend on avoidance behaviour and availability of the fish.

Avoidance

Evidence from towed gear studies suggests that small-mesh cod ends are avoided more than large-mesh cod ends, probably because the smaller mesh creates more drag and turbulence through the water (Wileman et al., 1996). Mesh size and colour influence the ability of fish to visually detect and avoid gill nets (Engås and Løkkeborg, 1994). Such avoidance, particularly that of towed gears, could be length dependent. The size of a bait on a hook could cause the strike probability (i.e. the probability that a fish tries to take the bait) to be length dependent, and the number, size and type of

fish already in a trap/pot may cause ingress to be size dependent.

Availability

The localized concentrations of fish that are available to the gear may have length distributions that differ from that of the entire population. Further, in the case of passive gears (e.g. gill nets, hooks and traps), availability is influenced by swimming speed (Rudstam et al., 1984), and this may be length dependent.

Comparative selectivity data contain no information about avoidance or availability and such knowledge must be obtained elsewhere. Consequently, only $r(l)$ can be estimated from a comparative selectivity study. Meaningful estimates of $s(l)$ (and $a(l)$) can only be obtained from $r(l)$ if independent knowledge of fish avoidance and the representativeness of the localized fish concentration is available. (Note that although comparative fishing experiments contain no information on avoidance of the (combined) gear, they do provide information on relative avoidance; that is, on whether one gear variant is avoided more than another. This is discussed later.)

The problems of estimating the population-selection curve, $s(l)$, are generally acknowledged in fish stock assessment (Anon., 1996). For example, in Atlantic Canada the retention curves, $r(l)$, of commercial trawls have been extensively modelled and estimated from various selectivity experiments. However, these retention curves are used only to a partial extent (e.g. to specify the smallest length at which retention is 'effectively' 100%, say), and it is usually accepted that the form of $s(l)$ is largely unknown because insufficient auxiliary information on fish behaviour exists to derive $s(l)$ from $r(l)$. In assessment meetings, several different forms of $s(l)$ are often considered. Recently, however, Myers and Hoenig (1997) used tagging studies to estimate the population selectivity for Atlantic cod and found little difference between these estimates and previously estimated retention curves. They concluded that the assumption of increased avoidance by larger cod was not supported by their analysis and was a major factor in the collapse of the cod stocks in Eastern Canada.

Comparative selectivity studies provide data for estimating only $r(l)$, and this review will concentrate on methods for doing just this. Henceforth, the term 'selection curve' is taken to be the contact-selection (i.e. retention) curve, $r(l)$, unless otherwise stated.

Fishing gears and their (contact-) selection curves

Towed gears

Towed gears essentially operate by surrounding and/or overtaking fish with netting, and fish must pass through the netting to escape. In most fish species, the vast majority of escape attempts occur in the cod end at the aft part of the gear (Fernö and Olsen, 1994), and hence selectivity is determined primarily by the size and shape of the mesh openings in the cod end, with larger fish being more likely to be retained than smaller fish. Towed gear selection curves are therefore assumed to be monotone nondecreasing functions of fish length with upper asymptote of unity. They are often summarized by the length of 50% retention (l_{50}) and the selection range (SR), the difference between the lengths of 75% and 25% retention. (In some species of Crustacea, escape has been observed in the main body of the trawl (Wileman et al., 1996) and hence cod-end selectivity may not provide an entire description of selectivity of the gear.)

The most commonly used mathematical description of towed gear selection curves is the logistic curve (Figure 1). This is specified by two parameters, a and b, and has the equation:

$$r(l) = \frac{\exp(a + bl)}{1 + \exp(a + bl)}. \tag{3}$$

The objective of the gear analyst is to estimate the parameters a and b, from which estimates of $l_{50} = -a/b$ and $SR = (2\log_e(3))/b$ are obtained (Wileman et al., 1996).

The logistic selection curve is symmetric about l_{50}. However, often an asymmetric selection curve is required. The extreme value and negative extreme value curves are two parameter curves with fixed amounts of asymmetry (Wileman et al., 1996). The Richards selection curve achieves varying degrees of asymmetry through use of an additional asymmetry parameter, δ, in the form:

$$r(l) = \left(\frac{\exp(a + bl)}{1 + \exp(a + bl)}\right)^{1/\delta}. \tag{3a}$$

When $\delta > 1$, the curve has a longer tail to the left of l_{50}, and when $0 < \delta < 1$ it has longer tail to the right (Figure 1). When $\delta = 1$ it reduces to the (symmetric) logistic curve.

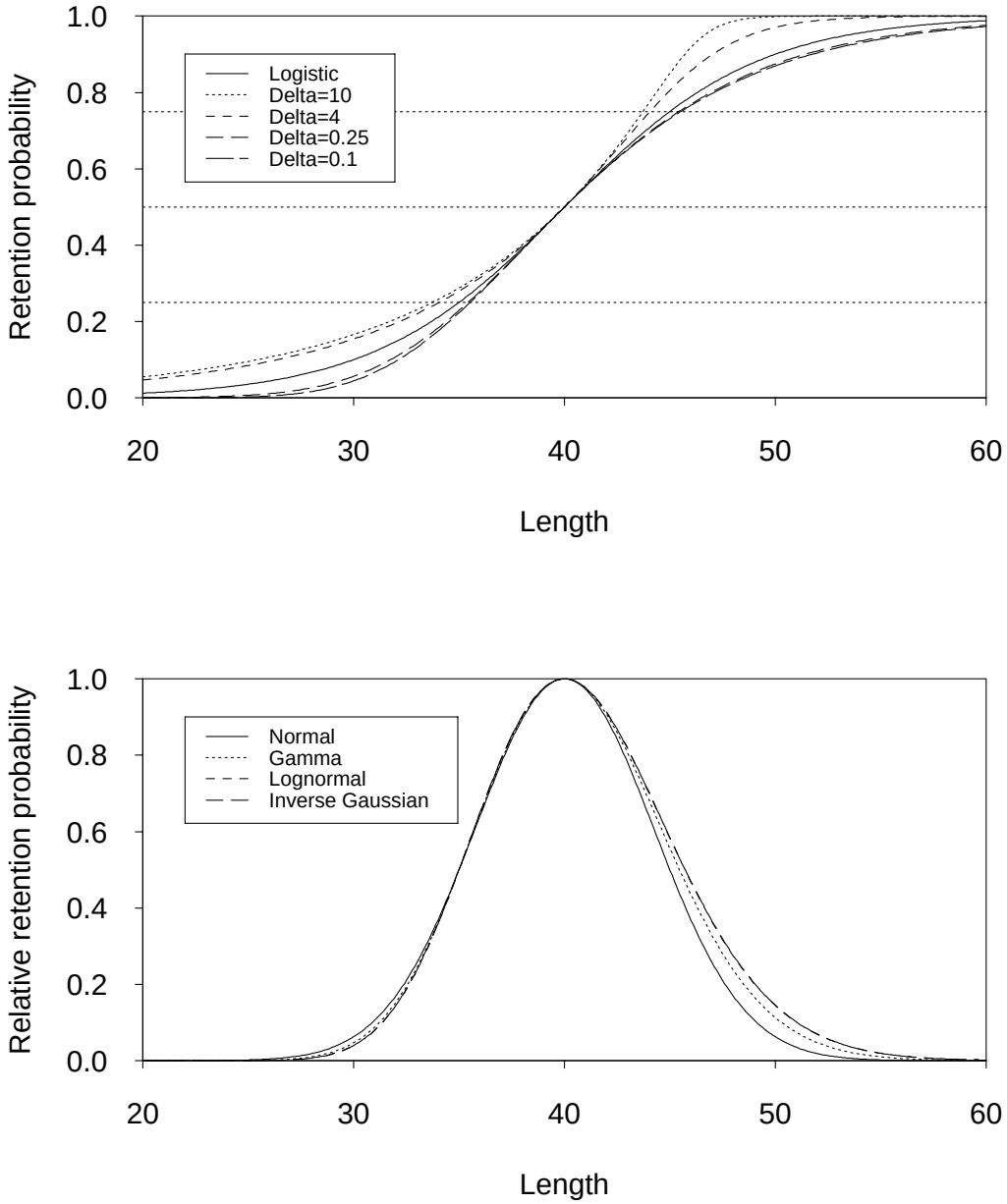


Figure 1. Top figure shows examples of monotone selection curves, including the logistic (solid curve) and four Richards selection curves corresponding to differing values of the asymmetry parameter, δ (dots, $\delta = 10$; short dashes, $\delta = 4$; medium dashes, $\delta = 0.25$; long dashes, $\delta = 0.1$). Parameters a and b of the curves (see text) were chosen to give a 50% retention length of 40 and selection range of 10. Bottom figure shows four unimodal selection curves, each with maximum height of unity at length 40, and height of 0.5 at length 35. The lognormal and inverse Gaussian are very similar and are indistinguisable on the figure, appearing as one line with a long dash pattern; solid curve, normal; dots, gamma.

Traps and pots

Traps and pots typically have openings which permit easy entry of fish (or crustaceans or molluscs) but make escape difficult. When escape from the trap opening is prohibited, then size selectivity is determined by the ability of the animal to pass through the mesh (say) of the trap or to exit through purpose-built escape gaps (if included). The selection curve will therefore be similar in shape to that of towed gear.

Gill nets

Gill nets (and entangling nets) are passive walls of netting that operate by wedging or entangling fish that swim into them (von Brandt, 1984). Sufficiently small fish will pass through the meshes, and sufficiently large fish will not penetrate far enough into the mesh to become wedged but may be snagged by a rough body part, or become entangled if the net is loosely rigged (as is the case with trammel nets).

The selection curve is often assumed to be bell shaped (i.e. normal or Gaussian) (Figure 1). This curve is summarized by two parameters, μ and σ, which measure its location and spread, and has equation:

$$r(l) \propto \exp\left(-\frac{(l-\mu)^2}{2\sigma^2}\right). \qquad (4)$$

Here, $r(l)$ is a relative retention probability because the constant of proportionality is unknown. However, without loss of generality ('General model', below), it can be assumed that gillnet selection curves have maximum height of unity.

The ability of gill nets, and particularly trammel nets, to capture large fish by snagging or entanglement implies that a selection curve that is skewed (Figure 1) or multimodal may be more appropriate than the normal curve (Hamley, 1975). The gamma, lognormal and inverse Gaussian curves are examples of unimodal skewed curves, all having a longer tail on the right than the normal (Figure 1). In practice, there is typically very little difference between the lognormal and inverse Gaussian curves (Cheng and Amin, 1981). A multimodal selection curve may be specified as the combination of two or more unimodal curves, a common example being the binormal combination of two normal curves (Hovgård, 1996).

Hooks

The usual commercial application of hooks is on a longline – a long length of line with leaders and baited hooks or lures attached at regularly spaced intervals. Very little is known about hook selection (Ralston, 1990), but it is believed that hooks have very broad selection curves (Pope et al., 1975). The typical scarcity of large fish in a commercial fishery makes it difficult to estimate the right-hand limb of the curve and consequently both monotone and unimodal curves have been proposed.

General model

Consider a comparative selectivity experiment with J gear variants. Let l denote the length class with midpoint l, let L be the number of length classes, and let n_{lj} be the number of length l fish caught in gear (variant) j.

We use the Poisson distribution to model the counts n_{lj}. This is the appropriate distribution for count data where the 'arrivals' of subjects are independent (Feller, 1968). In gear selectivity studies it is quite likely that arrivals will not be independent because of behavioural phenomena such as schooling. This results in overdispersion – variation in the observed fish counts greater than expected under the Poisson assumption. However, overdispersion does not generally cause any major problems and can readily be accounted for (McCullagh and Nelder, 1989).

Selection processes

Each n_{lj} depends on three processes: abundance, relative fishing intensity, and contact-selection, as follows.

Abundance

The fundamental assumption behind the general model is that the number of length l fish contacting the (combined) gear has a Poisson distribution, with mean λ_l say. The parameter λ_l can be thought of as the 'abundance' of length l fish contacting the gear.

Relative fishing intensity

The **relative fishing intensity** of length l fish in gear j, $p_j(l)$, is the probability that a fish of length l contacts gear j given that it contacts the combined gear. In most selectivity studies a fish will contact only one gear upon contact with the combined gear, and thus relative fishing intensities will sum to unity across gears, i.e., $\sum_j p_j(l) = 1$. (The possibility of multiple contacts may exist with gill nets and longlines. If the probability of multiple contacts is small or does not

vary greatly with fish length then it can reasonably be subsumed by the model by considering the fish to be once again available to the combined gear.) The one exception is the covered cod end, a special case we return to later.

In the simplest case, the relative fishing intensities will be length independent and proportional to the effort with which the gear is fished. Whence, if the gears are fished with equal effort, then $p_j(l) = 1/J$. However, a flexible $p_j(l)$ allows us to model **relative avoidance** where fish avoid some gears more than others, or conversely **fishing power** where some gears contact more fish than others.

Given that the number of contacts to the combined gear is Poisson distributed with mean λ_l, the number of length l fish contacting gear j is also Poisson distributed with mean $p_j(l)\lambda_l$ (Feller, 1968). This is just proportion $p_j(l)$ of the expected number λ_l contacting the combined gear.

Contact-selection

Let $r_j(l)$ be the contact-selection curve for gear j. The number of length l fish caught in gear j, n_{lj}, is then Poisson distributed (Feller, 1968) with mean $p_j(l)\lambda_l r_j(l)$; i.e. the expected number of contacts to gear j, $p_j(l)\lambda_l$, multiplied by the selection probability $r_j(l)$. This will be denoted

$$n_{lj} \sim Pois\left(p_j(l)\lambda_l r_j(l)\right). \qquad (5)$$

In practice, the contact-selection curves, $r_j(l)$, and the relative fishing intensities, $p_j(l)$, are confounded. That is, they cannot both be estimated from the data and specific assumptions must be made about either $r_j(l)$ or $p_j(l)$. Usually, it is assumed that the relative fishing intensities are constants, p_j, that do not depend on length, giving

$$n_{lj} \sim Pois\left(p_j\lambda_l r_j(l)\right). \qquad (6)$$

This is the general model for analysing data from all comparative selection studies.

There are three special cases worth noting.

1. In gillnet and hook studies, the maximum height of the selection curve may depend on the gear variant. However, this is confounded with p_j and hence it is enough to model relative selection curves (each with unit height) and allow the p_j to quantify the relative heights of the selection curves.
2. When there are only two gears, an experimental gear and one that prohibits all escapes, the selection curve and fishing intensity of the experimental gear can be denoted by $r(l)$ and p, respectively. The selection curve and fishing intensity of the escape-proof gear are then unity and $1-p$, respectively.
3. With the covered cod-end method, the two gear variants to be considered are the cod end and the cover. The relative fishing intensity of the cod end is 1 because all fish contacting the combined gear (cod end and cover) must first contact the cod end. Hence the catch of length l fish in the cod end is distributed $Pois(1 \times \lambda_l \times r(l)) = Pois(\lambda_l r(l))$. The relative fishing intensity of the cover is (by definition) the probability that a fish contacting the cod end escapes through to the cover, $1 - r(l)$. If the cover retains all cod-end escapees, the catch of length l fish in the cover is distributed $Pois\{[1 - r(l)] \times \lambda_l \times 1\} = Pois\{\lambda_l[1-r(l)]\}$. The more intuitive way of obtaining this result is to consider the cod end and cover to fish with equal fishing intensity and to have selection curves $r(l)$ and $1 - r(l)$, respectively. However, the strength of the general model is that it can model the effect of a cover that does not retain the smallest fish (the catch in the cover then being $Pois\{\lambda_l[1 - r(l)c(l)]\}$ where $c(l)$ is the retention curve of the cover), and it can easily incorporate the effect of subsampling from the cod-end or cover catch ('Subsampling', page 97).

Given this general framework, it is necessary to make assumptions and/or inferences about the form of p_j, λ_l and $r_j(l)$ that are appropriate for a particular study. For example:

1. If the gears are fished with equal effort, should the relative fishing intensities, p_j, be assumed equal, proportional to (some function of) mesh size, or should no assumptions be made? (detailed discussion, Hamley, 1975).
2. Is it reasonable to partially specify λ_l by postulating a form for the length distribution of fish contacting the gear? For example, can a normal, lognormal, or bimodal length distribution be assumed? Or should no assumptions on λ_l be made?
3. In a towed gear or trap study, is retention adequately modelled by a logistic curve or should a Richards curve be used? In a gillnet study, are the retention curves normal, gamma, lognormal, or bimodal?

The appropriate choices are made by fitting different models (i.e. different forms of p_j, λ_l, and $r_j(l)$) and by

comparing the fits using residual plots and goodness-of-fit statistics.

Windows and grids in towed gear

Modifications such as square-mesh panels (in the upper panel of the trawl extension piece) and rigid sorting grids (placed in front of the cod end) are receiving considerable attention by gear technologists. These devices are particularly effective with regard to reducing bycatch in towed trawls because they can take advantage of the different behaviour of animals entering the trawl. For example, shrimp tend to remain in the lower part of the trawl whereas fish will tend to be higher. The bycatch of fish in a shrimp trawl can therefore be reduced by including an escape window in the upper part of the trawl extension, possibly with the inclusion of a grid to herd fish towards the window (Isaksen and Valdemarsen, 1994).

Applying the general model of selectivity to a towed gear having windows or grids is similar to the application to covered cod-end gear (above). Each gear variant corresponds to a 'compartment' of the trawl in which fish are caught, and it is necessary to specify the catch rate for each. For example, the compartments of the modified shrimp trawl used by Tokai et al. (1996) were a small-mesh cover put over an escape window, and a small-mesh cod end. Fish were herded through the escape window by a stainless steel grid placed in the front of the cod end. In that study, Tokai et al. (1996) considered three equations to specify the catches in the cover over the escape window, one of which was $Pois([1-\delta]g(l)+\delta)$ where $g(l)$ is the probability that a length l fish is 'retained' by the grid (that is, is unable to pass through the grid into the cod end). Tokai et al. explain that this catch rate can be interpreted as the probability (δ) that the fish goes directly through the escape window without contacting the grid, plus the probability that it contacts the grid ($1-\delta$) and is unable to pass through it (which is given by the product of the probability of grid contact $1-\delta$ and probability of 'retention' by the grid given contact $g(l)$).

Subsampling

So far, it has been assumed that all the fish caught are measured. In practice, the catch is often subsampled, particularly in towed gear studies, where time is limited between deployments of the gear.

Subsampling is incorporated in the general model as follows. Let q_j be the subsampling fraction for gear j. This can be thought of as the probability that a fish retained by the gear is measured. The number of length l fish measured in gear j is then Poisson distributed with mean $p_j\lambda_l r_j(l)q_j$. For gear j, the subsampling fraction, q_j, is typically estimated by the ratio of the volume (or weight) of measured fish to the volume (or weight) of fish caught in that gear. The q_j's are then assumed known, and all the estimation and inference procedures that follow are modified accordingly. See Millar (1994) for full details.

Fitting the general model

The log-likelihood function

Maximum likelihood is used for all parameter estimation and inference. This guarantees known properties, including approximate unbiasedness and mimimum variance of parameter estimates (Cadigan and Millar, 1992). From Expression 6, the log-likelihood of the n_{lj} is (to within a constant that does not depend on any parameters)

$$\sum_l \sum_j \left\{ n_{lj} \log_e[p_j\lambda_l r_j(l)] - p_j\lambda_l r_j(l) \right\} \qquad (7)$$

which is a function of the unknown parameters $p_j,\ j=1,\ldots,J,\ \lambda_l,\ l=1,\ldots,L$, and the parameters defining the selection curves. The maximum likelihood estimates of the parameters are those values maximizing (7). At first sight, this maximization problem appears to be extremely difficult because of the many parameters to be considered. Typically there would be $J-1$ fishing intensities (because they sum to unity), L abundances, and between 2 (logistic curve) and $5J$ (binormal curve for each of J gillnets) selection parameters. One could ignore this consideration and apply a numerical optimizer to the likelihood. However, often this will not work owing to numerical instability and identifiability (i.e. overparametrization) problems. Fortunately, however, there are several ways in which the problem can be simplified.

Specifying the population length distribution

One way of reducing the number of parameters is to assume a parametric form for the length distribution of the fished population $\lambda_l, l=1,\ldots,L$. For example, if the population length distribution is normally distributed with mean θ and standard deviation τ then $\lambda_l = N\cdot\exp\left[-(l-\theta)^2/2\tau^2\right]$, where N is a scaling parameter that measures 'overall abundance'. The number of parameters is thus reduced by $L-3$.

This is perhaps an area where more research should be invested. For example, it is not known how sensitive this approach is to mis-specification of the shape of the population length distribution. In practice, it will generally be difficult to specify the length distribution with much certainty.

In a gillnet study, Boy and Crivelli (1988) assumed that the population length distribution was a mixture of age classes for which they had length-at-age data, and estimated the proportion of each age class simultaneously with the gillnet retention probabilities. However, their estimation methodology was *ad hoc* and selection curves were not formally modelled.

Geometric similarity

When fishing with several gear variants that differ only in mesh size (or size of escape gap or hook), such as in a gillnet study, the number of parameters can be greatly reduced by assuming that selection curves $r_j(l)$ differ in a systematic way depending only on the mesh size, m_j, say.

The most widely used assumption is the principle of geometric similarity (Baranov, 1914, 1948), which assumes that selection depends only on the relative geometry of the mesh and the fish: "Since all meshes are geometrically similar and all fish of the same species (within a reasonable size range) are also geometrically similar, the selection curves for different mesh sizes must be similar" (quoted, in translation, from Wulff, 1986, p. 101). Specifically, the principle implies:

$$r_j(l) = r_{j'}\left(\frac{m_{j'}}{m_j}l\right). \tag{8}$$

For example, suppose gill net j' has double the mesh size of gillnet j, so that $m_{j'}/m_j = 2$. Equation (8) then implies that the retention probability of a length l fish by gill net j is the same as the retention probability of a length $2l$ fish by gill net j'. In particular, if gill net j has mode (maximum retention) at μ and spread σ, then gill net j' has mode at 2μ and spread 2σ. Thus, in a gillnet study with J gears, if each is assumed to have a normal retention curve, then geometric similarity reduces the number of selection curve parameters from $2J$ to 2. Geometric similarity can also be used in towed gear studies, where doubling cod-end mesh size implies doubling the l_{50} and SR of the cod end.

One consequence of geometric similarity is a common maximum height of the selection curve for all gears. This is implicit for towed gear and trap studies where all selection curves have a maximum height of unity, but in gillnet studies, Hamley (1975) considers this the major objection to the principle of geometric similarity. However, Hamley's criticisms are directed against the population-selection curves, $s_j(l)$, and do not necessarily apply to the contact-selection curves, $r_j(l)$, modelled here. In particular, length-dependent swimming speed will affect the maximum heights of $s_j(l)$, but not $r_j(l)$. Here, we can assume a common height without loss of generality, because differences in avoidance of different mesh sizes are incorporated into the relative fishing intensities p_j.

One argument against the principle of geometric similarity of $r_j(l)$ is that the body shape of fish may change as they grow older and girths may therefore not be proportional to lengths (Hamley, 1975). Undoubtedly, geometrical similarity is an oversimplification, but it does provide a reasonable a priori way of reducing the number of selection parameters to be estimated. In particular, specifying the selection curve for the smallest gear (say) determines the selection curves for all the gears. Clearly, however, geometric similarity must not be assumed if the data provide evidence to the contrary.

Log-linear models

For some gillnet and hook studies, it is possible to take advantage of special structure in the selection curves to estimate the parameters using standard statistical software. Let $\eta_{lj} = E(n_{lj}) = p_j\lambda_l r_j(l)$ denote the expected catch of length l fish in gear j. Then,

$$\log_e\left(\eta_{lj}\right) = \log_e\left(p_j\right) + \log_e\left(\lambda_l\right) + \log_e\left(r_j(l)\right). \tag{9}$$

For several of the selection curves used in gillnet and hook studies, $\log_e(\eta_{lj})$ can be expressed in log-linear form (Millar and Holst, 1997),

$$\log_e\left(\eta_{lj}\right) = \sum_i \beta_i \cdot f_i(m_j, l), \tag{10}$$

where $f_i(m_j, l)$ are known functions of m_j and l and do not depend on any unknown parameters. This is now a standard log-linear model (McCullagh and Nelder, 1989) and can be fitted using the log-linear modelling capabilities of many statistical software packages. Millar and Holst (1997) show how fits can be achieved in as little as three lines of program code, and how the parameters of the selection curves are obtained from the estimated β_i. The selection curves

that can be fitted using log-linear modelling include the normal, gamma, lognormal and inverse Gaussian.

For example, suppose that (i) the gill nets have equal fishing intensity, (ii) the form of the population length distribution is not assumed, and (iii) the retention curves are assumed to be normal shaped and to observe geometric similarity, with mode $\mu_j = k_1 m_j$ and spread $\sigma_j = k_2 m_j$ proportional to mesh size. Millar and Holst (1997) show that $log_e(\eta_{lj})$ can be written in the log-linear form

$$\log(\eta_{lj}) = \text{factor}(l) + \beta_1 \cdot f_1(m_j, l) + \beta_2 \cdot f_2(m_j, l) \qquad (11)$$

where $f_1(m_j, l) = l/m_j$ and $f_2(m_j, l) = l^2/m_j^2$. The notation factor(l) denotes that length class is treated as a factor, and hence any terms in $\log(\eta_{lj})$ that are constants or functions of l alone are absorbed in the fit of the factor (Milar and Holst, 1997). Parameters k_1 and k_2 are then estimated by $\hat{k}_1 = \hat{\beta}_1/(-2\hat{\beta}_2)$, $\hat{k}_2 = (-2\hat{\beta}_2)^{-\frac{1}{2}}$. Table 1 gives the log-linear form for various other selection curves.

The log-linear form of $\log(\eta_{lj})$ has long been recognized in the selectivity literature for hook and gillnet selectivity analysis (Holt, 1963). However, prior to the general model developed here, previous selectivity analyses that have exploited this structure have been *ad hoc*, have not modelled the data as counts, and could not fit to more than two gear variants.

SELECT

The SELECT method eliminates the λ_l parameters from the maximization problem altogether. It does so by working with the *proportions* of the total catch (for each length class) taken by each gear. Let $n_{l+} = \sum_j n_{lj}$ be the total catch of length class l by all J gears combined and let $y_{lj} = n_{lj}/n_{l+}$ be the proportion of the length class l catch taken in gear j. From the general model, Expression 6, it can be shown (Feller, 1968) that the proportions, y_{lj}, $j = 1, \ldots, J$, have a multinomial distribution with n_{l+} trials and probabilities

$$\phi_{lj} = E(y_{lj}) = \frac{p_j \lambda_l r_j(l)}{\sum_j p_j \lambda_l r_j(l)} = \frac{p_j r_j(l)}{\sum_j p_j r_j(l)}, \quad j = 1, \ldots, J \qquad (12)$$

where E is the expected value.

The important thing to note is that the probabilities ϕ_{lj} do not depend on λ_l. The log-likelihood for the data y_{lj}, $l = 1, \ldots, L$, $j = 1, \ldots, J$, is then:

$$\sum_l \sum_j n_{lj} \log_e(\phi_{lj}). \qquad (13)$$

Maximizing the log-likelihood (13) gives the same selectivity parameter estimates (p_j, $j = 1, \ldots, J$, and the parameters defining the selection curves) as maximizing the log-likelihood (7). Moreover, all inferences regarding these parameters will be identical (McCullagh and Nelder, 1989; Millar, 1992). However, using SELECT, the λ_l do not need to be estimated at all.

The SELECT approach reduces the number of parameters dramatically. When fitting a logistic selection curve to covered cod-end data, there are only two parameters to estimate, a and b. This increases to three in trouser trawl experiments, when relative fishing intensity, p, is also estimated. In experiments with several sizes of hook or gillnet meshes, the number of parameters depends inversely on the number of model assumptions. For example, assuming a gamma selection curve, geometric similarity, and equal fishing intensities, the SELECT approach has only two parameters to estimate: the parameters of the selection curve for the smallest size hook or mesh. SELECT generally requires a numerical maximization routine. However, in the special case of covered cod-end data, SELECT corresponds to a standard binomial regression, and parameter estimates for a logistic selection curve can be obtained using the generalized linear modelling facilities of standard statistical software.

When there are only two gears, working with proportions has the additional advantage of reducing the data to a single proportion for each length class. This allows the data to be visually inspected by a plot of proportion versus length. For example, in a covered cod-end experiment, it is usual to plot the proportion retained in the cod end against length. This plot suggests suitable forms of the selection curve. In a trouser trawl experiment, the proportion retained in the experimental cod end is plotted against length (Millar, 1992; Millar and Walsh, 1992). This is particularly useful for indicating the relative fishing intensities of the two cod ends.

Table 1. Models for normal, lognormal, gamma, inverse Gaussian, and binormal selection curves. The first normal selection curve model, Normal loc., assumes fixed spread and is not geometrically similar. All others are geometrically similar. As specified, all except the binormal are of unit height. (Unit height can be assumed without loss of generality because relative fishing intensity is modelled separately.) In the case of the binormal, the height is not specified because it can not be obtained explicitly and is not required for fitting purposes. The right-hand column gives the last two terms in the log-linear model

Model	Selection curve	$[\beta_1]\cdot\{f_1(m_j,l)\}+[\beta_2]\cdot\{f_2(m_j,l)\}$
Normal loc. k,σ	$\exp\left(-\frac{(l-k\cdot m_j)^2}{2\sigma^2}\right)$	$\left[\frac{k}{\sigma^2}\right]\cdot\{l\cdot m_j\}+\left[-\frac{k^2}{2\sigma^2}\right]\cdot\{m_j^2\}$
Normal k_1,k_2	$\exp\left(-\frac{(l-k_1\cdot m_j)^2}{2k_2^2\cdot m_j^2}\right)$	$\left[\frac{k_1}{k_2^2}\right]\cdot\left\{\frac{l}{m_j}\right\}+\left[-\frac{1}{2k_2^2}\right]\cdot\left\{\left(\frac{l}{m_j}\right)^2\right\}$
Lognormal μ,σ	$\frac{m_j}{l\cdot m_1}\exp\left(\mu-\frac{\sigma^2}{2}-\frac{\left(\log(l)-\mu-\log\left(\frac{m_j}{m_1}\right)\right)^2}{2\sigma^2}\right)$	$\left[\frac{1}{\sigma^2}\right]\cdot\left\{\log(l)\cdot\log\left(\frac{m_j}{m_1}\right)-\frac{1}{2}\log^2\left(\frac{m_j}{m_1}\right)\right\}$ $+\left[1-\frac{\mu}{\sigma^2}\right]\cdot\left\{\log\left(\frac{m_j}{m_1}\right)\right\}$
Gamma α,k	$\left(\frac{l}{(\alpha-1)\cdot k\cdot m_j}\right)^{\alpha-1}\exp\left(\alpha-1-\frac{l}{k\cdot m_j}\right)$	$[\alpha-1]\cdot\left\{\log\left(\frac{l}{m_j}\right)\right\}+\left[-\frac{1}{k}\right]\cdot\left\{\frac{l}{m_j}\right\}$
Inverse Gaussian k_1,k_2	$(k/l)^{3/2}\exp\left(\frac{-k_2}{2k_1^2\cdot m_j}\left(\frac{(l-k_1\cdot m_j)^2}{l}-\frac{(k-k_1\cdot m_j)^2}{k}\right)\right)$ where $k=\frac{k_1^2}{k_2}\cdot m_j\left(\sqrt{\frac{9}{4}+\frac{k_2^2}{k_1^2}}-\frac{3}{2}\right)$	$\left[-\frac{k_2}{2k_1^2}\right]\cdot\left\{\frac{l}{m_j}\right\}+\left[-\frac{k_2}{2}\right]\cdot\left\{\frac{m_j}{l}\right\}$ $+\text{offset}\left(\frac{3}{2}\log(m_j)\right)$
Binormal k_1,k_2,k_3,k_4,c	$\exp\left(-\frac{(l-k_1\cdot m_j)^2}{2k_2^2\cdot m_j^2}\right)+c\exp\left(-\frac{(l-k_3\cdot m_j)^2}{2k_4^2\cdot m_j^2}\right)$	Not applicable — use SELECT

Available selectivity software

Software for selectivity analysis has been written in a variety of computer packages and languages. A number of these are freely available from

http://www.scitec.auckland.ac.nz/∼greebie/selectware/code.html

This URL contains Splus code for covered cod-end, trouser trawl, and gillnet (log-linear models only; see Millar and Holst, 1997) analyses; FORTRAN (Millar and Cadigan, 1991) and SAS code (Millar, 1993a) for trouser trawl; and Excel spreadsheets (kindly provided by Prof. Tadashi Tokai, Tokyo University of Fisheries; see Tokai, 1997; Tokai and Mitushashi, 1998).

For fully documented and supported software, see the commercial product described at

http://www.constat.dk

Inference

Assessment of fit, model selection, and degrees of freedom

The fit of a selectivity model is assessed by using a goodness-of-fit statistic *and* through inspection of residuals.

There is some choice of goodness-of-fit statistic and the type of residual to use when assessing fits to count data (McCullagh and Nelder, 1989). Here, the model deviance was used as a goodness-of-fit statistic, calculated as twice the difference between the Poisson log-likelihoods of the 'full' model and the current model. Correspondingly, deviance residuals $D_{lj}, l = 1, \ldots, L, \ j = 1, \ldots, J$ have been used throughout. (D_{lJ} is the signed square root of the contribution to the model deviance from observing n_{lj}.)

When only two gear variants are deployed then, for each length class l, it is useful to combine the deviance residuals D_{l1} and D_{l2} to get a deviance residual for the length class, given by

$$D_l = sign(n_{l1} - \hat{\eta}_{l1})\sqrt{D_{l1}^2 + D_{l2}^2}. \qquad (14)$$

This corresponds to the deviance residual obtained from the binomial log-likelihood (McCullagh and Nelder, 1989, p. 118).

The model deviance has an approximate χ^2 (chi-square) distribution when the model is correct and the expected catches, η_{lj} are all sufficiently large (> 3, say). The degrees of freedom (d.f.) of the χ^2 distribution is usually taken to be the number of nonzero $\hat{\eta}_{lj}$ minus the number of parameters. If using the SELECT method of fitting, it is important to be aware that the SELECT method is *implicitly* estimating the population length distribution parameters, λ_l, and hence it is necessary to include these as parameters when calculating the d.f.

When the $\hat{\eta}_{lj}$ are small, the model deviance and d.f should be recalculated using a subset of length classes with reasonable catches. For hypothesis testing between nested models the model deviance is less sensitive to small η_{lj}. When the simpler model is correct, then the difference in model deviance of the two models is approximately a χ^2 distribution with d.f. given by the difference between the number of model parameters.

Overdispersion

Lack of fit (as indicated by a large model deviance) does not necessarily imply that the fitted selection curve is not a good model of the selectivity of the gear. If a plot of residuals versus length shows no clear structure, then the lack of fit might result from overdispersion (McCullagh and Nelder, 1989), a failure of the assumption that fish behave independently.

Overdispersion has little effect on parameter estimation. However, it does affect parameter standard errors and hypothesis tests. The degree of overdispersion is measured by the dispersion parameter, which can be estimated by the ratio of the model deviance to its d.f., provided that the length classes used for this calculation are not too sparse. To correct inferences for overdispersion, all test statistics based on the difference between model deviances should be divided by the dispersion parameter (calculated from the fuller model), and all standard errors should be multiplied by the square root of the dispersion parameter.

Parameter standard errors

Parameter standard errors are obtained by standard maximum likelihood theory (McCullagh and Nelder, 1989; Collett, 1991; Millar and Walsh, 1992; Wileman et al., 1996). For the log-linear models and those SELECT models that can be fitted using standard statistical software, standard errors are automatically produced as part of the software output.

The parameters used in the model may not be the parameters of direct interest (e.g. l_{50} and *SR* in a towed

gear study). The standard errors for these derived parameters can be obtained using the delta method (Millar and Walsh, 1992; Wileman et al., 1996). Alternatively, in some cases it may be possible to (uniquely) express the model explicitly in terms of the parameters of interest and hence obtain the standard errors directly.

Examples

Entangling net catches of plaice

The data (Table 2) are catches of plaice from an experiment conducted during summer and autumn 1994 (Holst and Moth-Poulsen, 1995). Gangs consisting of six trammel nets (three wall nets) of differing inner mesh size were fished, with the order of each inner mesh size being chosen at random each time. The average measured inner mesh sizes were 9.83, 10.84, 11.94, 12.87, 13.97 and 15.07 cm. The outer walls were of 60 cm mesh and are presumed not to influence the selectivity of the inner meshes. Each net was 1000 inner meshes long and of common hanging ratio. Hence the length, and therefore fishing effort, of each net was approximately proportional to its mesh size.

Six families of selection curves were fitted to the data (Table 1, Figure 2). Geometric similarity is observed in five of these families, with both the modal length and spread of the curves increasing proportional to mesh size. The one exception is the normal location curve, where the modal length is proportional to mesh size, but the spread of the curve is fixed over mesh size. The five unimodal curves were fitted using the log-linear approach (via function *glm* in the Splus statistical software) and SELECT was used to fit the binormal (bimodal normal) curves. Relative fishing intensity was assumed to be proportional to mesh size, because effort was proportional to mesh size.

Model deviances were calculated using all length classes with nonzero catch, and again for only the 19 length classes with total catch of at least 18 (corresponding to an 'average' catch in each mesh of at least 3). In the latter case, the two-parameter unimodal selection curve fits have model deviance with d.f. equal to 93 ($19 \times 6 - 19 - 2$), and d.f. is 90 for the five-parameter binormal.

The binormal fit gives a drastic reduction in model deviance (Table 3) compared with the unimodal curves. There is a corresponding improvement in the plot of deviance residuals (Figure 3) and hence the binormal is by far the superior model. Nonetheless, its model deviance is substantially higher than its d.f. (166.3 on 90 d.f.) and the structure remaining in the deviance residuals does suggest some lack of fit.

The models were also fitted assuming that fishing power is proportional to mesh size; that is, with relative fishing intensity proportional to mesh size squared. The binormal fit is still the best (Table 3),

Table 2. Catches of plaice by multimono trammel nets (three wall nets), from Holst and Moth-Poulsen (1995). The outer walls were 60 cm mesh and inner mesh sizes were 9.83, 10.84, 11.94, 12.87, 13.97, and 15.07 cm. Fishing effort increased in proportion to inner mesh size because each of the nets was 1000 inner meshes long

Length	Mesh size (cm)					
(cm)	9.83	10.84	11.94	12.87	13.97	15.07
20	2	0	0	0	0	0
21	1	3	0	0	0	0
22	12	7	0	1	1	0
23	41	7	0	1	1	1
24	80	18	12	5	2	2
25	113	91	21	6	5	1
26	109	198	41	19	6	3
27	115	239	81	19	17	5
28	87	203	186	40	26	13
29	72	165	221	80	35	10
30	46	90	176	125	31	16
31	22	68	126	124	50	14
32	13	43	64	95	63	22
33	9	16	34	56	48	18
34	11	16	16	33	45	21
35	5	10	14	20	27	23
36	4	3	10	18	17	22
37	3	3	8	13	19	10
38	3	2	5	5	9	13
39	3	6	2	4	6	8
40	1	2	7	2	5	5
41	2	0	2	1	1	5
42	2	0	1	2	1	4
43	1	0	0	3	3	4
44	2	3	0	1	4	1
45	0	1	0	1	0	1
46	0	0	1	1	0	0
47	0	0	1	0	1	1
48	0	0	0	0	0	0
49	0	0	0	0	0	0
50	0	0	0	0	1	0
51	0	0	0	1	0	0
52	0	0	0	0	0	0
53	0	0	0	0	0	1
54	0	0	0	0	0	0
55	0	0	0	0	0	1

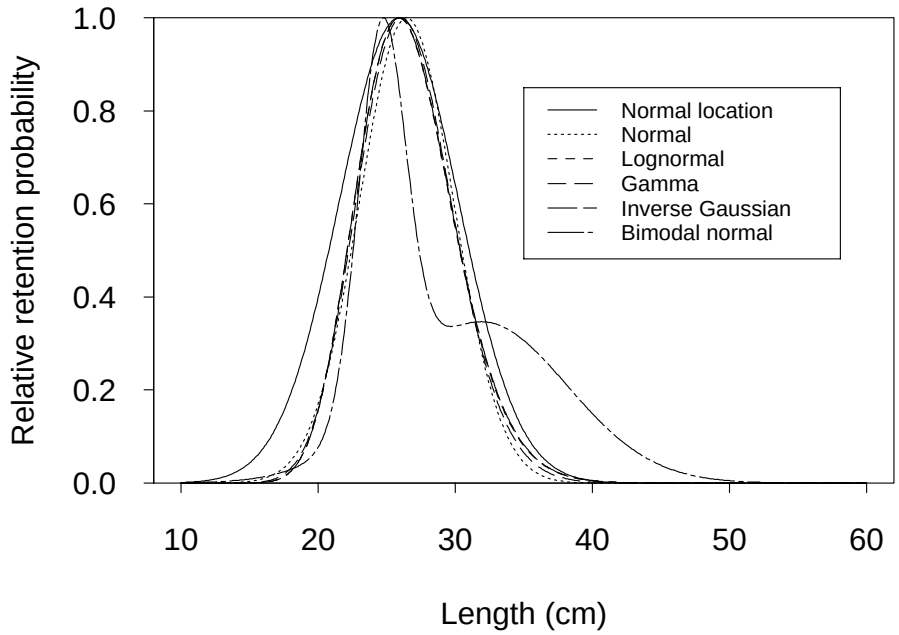


Figure 2. Estimated selection curves for the 9.83 cm (inner) mesh trammel net (Table 2).

Table 3. Fits to the plaice data of Table 2. The model deviance and degrees of freedom (d.f.) were calulated using the 19 length classes with a total catch of 18 or more. For comparison, the model deviances calculated using all 32 length classes with nonzero catch are also given (in square brackets) to indicate the inflation in d.f. caused by using sparse data. Parameters given are for the 9.83 cm mesh size ($m_1 = 9.83$)

Model	Equal fishing power		Fishing power $\propto$ mesh size	
	Parameters	Deviance: d.f.	Parameters	Deviance: d.f.
Normal loc.				
$\mu = k \cdot m_1$	$(\mu, \sigma) = (25.9, 4.35)$	604.6: 93 [783.3: 158]	$(\mu, \sigma) = (26.4, 4.41)$	600.5: 93 [776.3: 158]
Normal				
$\mu = k_1 \cdot m_1$	$(\mu, \sigma) = (26.5, 3.43)$	712.1: 93 [960.8: 158]	$(\mu, \sigma) = (26.9, 3.40)$	714.5: 93 [965.0: 158]
$\sigma^2 = k_2 \cdot m_1^2$				
Lognormal	$(\mu, \sigma) = (3.27, 0.134)$	530.5: 93 [685.1: 158]	$(\mu, \sigma) = (3.29, 0.134)$	530.5: 93 [685.1: 158]
Gamma				
$\beta = k \cdot m_1$	$(\alpha, \beta) = (57.8, 0.459)$	572.7: 93 [755.0: 158]	$(\alpha, \beta) = (58.8, 0.459)$	572.7: 93 [755.0: 158]
Inverse Gaussian				
$\mu = k_1 \cdot m_1$	$(\mu, \lambda) = (26.6, 1459)$	534.1: 93 [688.6: 158]	$(\mu, \lambda) = (27.1, 1484)$	533.7: 93 [687.6: 158]
$\lambda = k_2 \cdot m_1$				
Binormal				
$\mu_1 = k_1 \cdot m_1$				
$\sigma_1^2 = k_2 \cdot m_1^2$	(μ_1, σ_1) = (24.6, 1.72)	166.3: 90	(μ_1, σ_1) = (24.8, 1.72)	166.8: 90
$\mu_2 = k_3 \cdot m_1$	(μ_2, σ_2) = (31.9, 6.24)	[229.2: 155]	(μ_2, σ_2) = (33.1, 6.09)	[229.9: 155]
$\sigma_2^2 = k_4 \cdot m_1^2$	c = 0.42		c = 0.56	

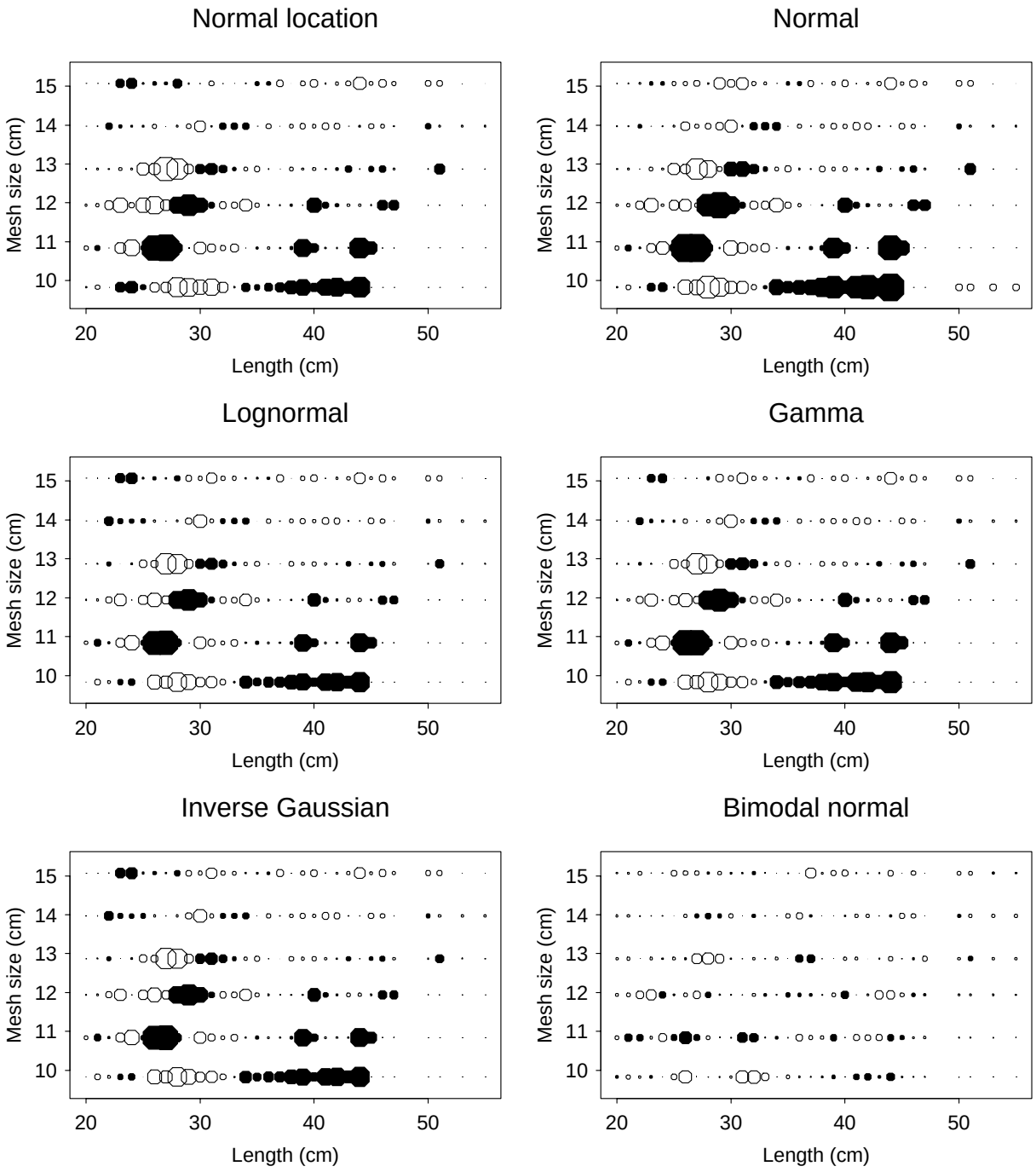


Figure 3. Deviance residuals resulting from the fits (Figure 2) to the trammel net data of Table 2. Empty and filled octagons correspond to negative and positive residuals, respectively, and the area of the octagon is proportional to the squared residual.

Table 4. Alternate-hauls haddock data from Table II of Pope et al. (1975). The experimental and control cod ends used 87 mm and 35 mm mesh respectively

Length (cm)	Number in control cod end	Number in experimental cod end	Proportion in experimental cod end
24	1	0	0.000
25	1	0	0.000
26	3	0	0.000
27	14	1	0.067
28	30	5	0.143
29	49	19	0.279
30	60	29	0.326
31	50	51	0.505
32	70	91	0.565
33	108	120	0.526
34	88	118	0.573
35	84	107	0.560
36	68	78	0.534
37	37	52	0.584
38	33	40	0.548
39	12	17	0.586
40	5	17	0.773
41	6	14	0.700
42	10	10	0.500
43	1	4	0.800
44	6	6	0.500
45	2	2	0.500
46	1	5	0.833
47	0	1	1.000

though its deviance is now a little worse than under the assumption of equal fishing power of the different-sized meshes.

Alternate haul catches of haddock

The data (Table 4) are alternate-haul haddock data from Table II of Pope et al. (1975). These data have been used previously to demonstrate application of the general selectivity model – see Millar (1992) or Wileman et al. (1996) for greater detail than given here.

We begin by fitting a logistic selection curve with the relative fishing intensity of the experimental cod-end p fixed at 0.5 so that it equals the relative fishing intensity of the small-mesh cod end, $1 - p$. The model deviance (Table 5) indicates a problem with the fitted model. The deviance residuals (Figure 4) show that it is lack of fit because the fitted model under-

Table 5. Maximum likelihood fits of a logistic selection curve to alternate-haul haddock data (Table 4). Parameter estimates are given for both the equal ($p = 0.5$) and unequal relative fishing efficiency assumptions. Model fit was assessed from Figure 4 and using deviance and d.f.'s calculated using only the 24 length classes with expected catch of greater than 3 in both the experimental and control cod ends. Deviance and d.f.'s calculated using all data are shown in square brackets, and are used to test $H_0 : p = 0.5$ (see section 'Overdispersion'). Values in parentheses are standard errors

	$p = 0.5$	p estimated
a	−36.3	−27.7 (5.8)
b	1.23	0.92 (0.20)
p	0.5	0.57 (0.02)
l_{25}	28.5	29.0 (0.3)
l_{50}	29.4	30.2 (0.4)
l_{75}	30.3	31.4 (0.6)
H_0: fit		
Deviance	32.5 [36.0]	10.4 [14.8]
d.f.	15 [22]	13 [21]
P value	0.006	0.66
$H_0 := 0.5$		
Deviance		21.2
d.f.		1
P value		< 0.000001

estimates the proportion of large fish retained in the experimental cod end.

Recall that p is the probability that a fish contacts the experimental cod end given that it contacts the combined (experimental and small-mesh) gear, and quantifies the difference in fishing power of the two cod ends. For the larger fish (those unable to escape the experimental cod end), the proportion caught in the experimental cod end has expected value equal to p. Table 4 and Figure 4 suggest that p should not be fixed at 0.5. Allowing the model to estimate p is a major improvement: the model deviance no longer indicates lack of fit, and the deviance residuals show no obvious remaining structure (Figure 4). It is appropriate to choose this model and complete the analysis by calculation of the standard errors of the selection parameters.

Covered cod-end haddock data

The data (Table 6) are total haddock catches from three hauls each of 20 min. duration and 60 min. duration (Clark, 1957). Here, we analyse only the total catch

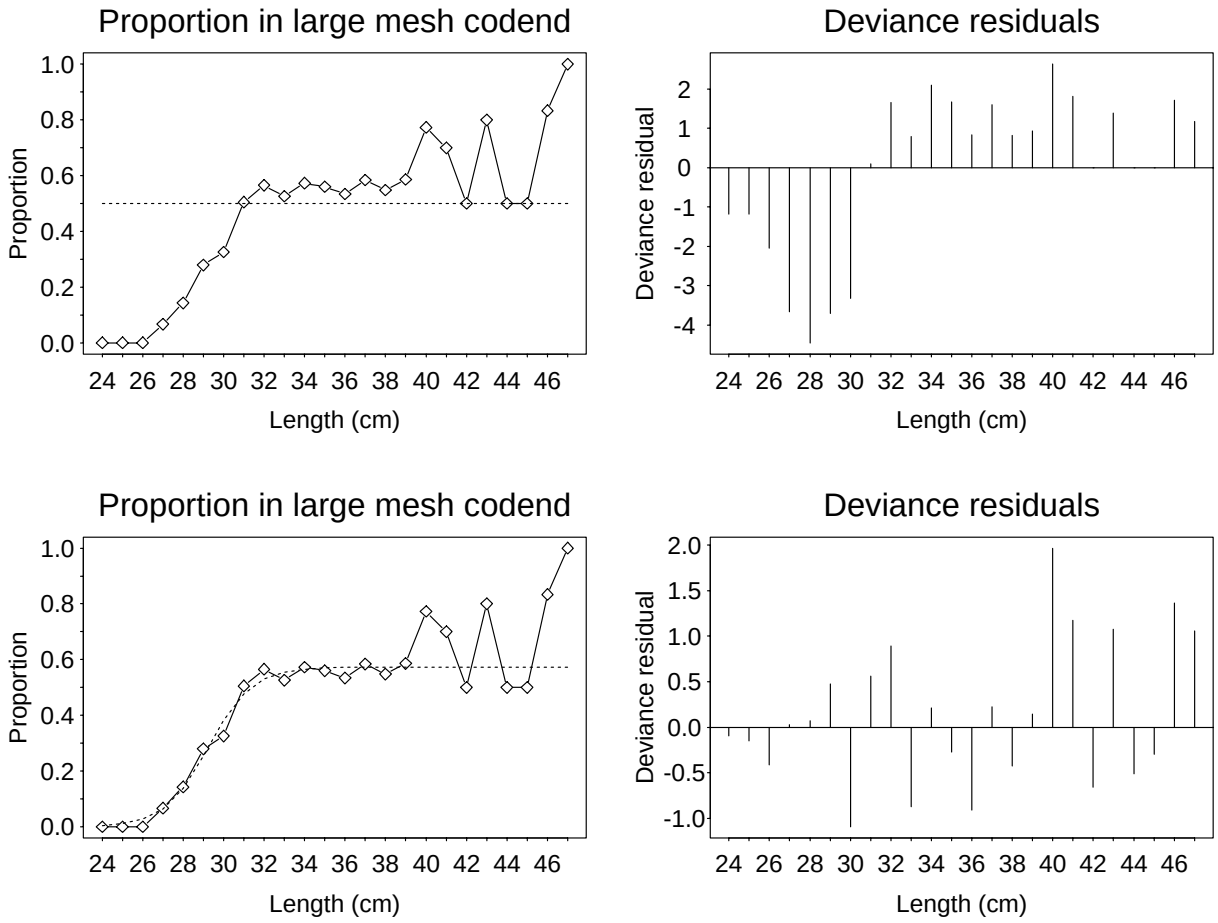


Figure 4. Fit of logistic selection curves to the alternate data of Table 4 and the resulting deviance residuals. The upper plots are from an equal fishing intensity ($p = 0.5$) fit. The estimated p case is given in the lower plots.

summed over the three hauls of each duration. (The hauls are analysed individually below.)

A logistic selection curve fits well (Table 7, Figure 5) to the 20 min. data, notwithstanding that there are noticeably more positive deviance residuals for lengths above 38 cm. However, this is not too severe and to some extent is expected, because when retention probabilities approach 1, then the observed retention rate is often exactly 1. It would be appropriate to stop with the logistic fit to the 20 min. data. For interest, however, the fit of a Richards curve is also given (Table 7). Consistent with the logistic being an adequate fit, the Richards curve does not provide a significant improvement (P value ≈ 0.3).

Analysis of the 60 min. data is more challenging. The model deviance of 36.5 on 20 d.f. suggests model inadequacy (P value ≈ 0.01), and the the deviance residuals (Figure 5) confirm that there is lack of fit. The logistic selection curve is clearly not acceptable for these data.

Two other analyses were performed. First, a Richards curve provided a significantly better fit (P value < 0.001) with a major reduction in model deviance (Table 7). The deviance residuals (Figure 5) are much improved, but do show a long run of negative residuals from 26.5 to 33.5 cm.

Secondly, the analysis was repeated with the length classes below 27.5 cm removed. This cut off was chosen because it appears to eliminate the long left-hand tail of the observed retention proportions while retaining as many length classes as possible. The logistic and Richards fits to the reduced data set were almost identical and provide adequate fit (Table 7, Figure 5).

Whether one prefers the Richards fit to the full data, or the logistic fit to the reduced data, depends

Table 6. Individual-haul catch data for haddock in three 20 min. and three 60 min. covered cod-end hauls with 113 mm mesh (C, cod end; V, cover). Data from Clark (1957)

Length (cm)	20 min. hauls								60 min. hauls							
							Totals								Totals	
	C	V	C	V	C	V	C	V	C	V	C	V	C	V	C	V
18.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
19.5	0	0	0	0	0	2	0	2	0	3	0	1	1	0	1	4
20.5	0	0	0	1	0	4	0	5	0	9	0	4	1	12	1	25
21.5	0	0	0	0	0	11	0	11	2	25	0	6	0	38	2	69
22.5	0	0	0	9	0	19	0	28	7	50	2	23	1	78	10	151
23.5	0	1	0	10	1	42	1	53	9	80	2	41	2	69	13	190
24.5	0	0	0	4	5	42	5	46	7	78	0	40	3	74	10	192
25.5	0	0	1	5	0	30	1	35	9	59	1	39	4	48	14	146
26.5	0	1	0	2	3	24	3	27	5	51	1	23	1	28	7	102
27.5	0	0	0	0	1	5	1	5	1	30	0	21	0	11	1	62
28.5	0	0	0	0	0	3	0	3	0	17	0	3	1	4	1	24
29.5	0	0	0	0	1	2	1	2	0	2	0	1	0	1	0	4
30.5	0	0	0	0	0	0	0	0	1	7	0	0	0	1	1	8
31.5	0	0	0	0	0	2	0	2	2	7	0	1	0	4	2	12
32.5	0	0	1	0	2	2	3	2	4	8	1	2	0	9	5	19
33.5	0	1	0	0	4	3	4	4	0	7	1	1	1	8	2	16
34.5	2	3	1	4	2	5	5	12	11	17	3	2	2	8	16	27
35.5	3	3	0	0	5	6	8	9	13	16	2	4	3	11	18	31
36.5	2	2	4	3	7	9	13	14	13	20	3	3	7	14	23	37
37.5	8	4	7	3	14	8	29	15	10	21	14	4	12	7	36	32
38.5	10	1	4	0	15	7	29	8	23	21	12	5	22	8	57	34
39.5	12	1	5	5	17	3	34	9	21	9	7	2	18	13	46	24
40.5	8	0	5	1	17	2	30	3	19	12	14	1	22	4	55	17
41.5	11	0	5	0	13	3	29	3	11	4	10	1	21	3	42	8
42.5	3	0	3	0	12	2	18	2	12	1	4	0	10	0	26	1
43.5	2	0	2	0	12	1	16	1	6	2	7	0	5	0	18	2
44.5	1	0	2	0	8	0	11	0	13	0	5	1	4	1	22	2
45.5	3	0	3	0	6	0	12	0	5	0	0	0	1	0	6	0
46.5	2	0	2	0	5	0	9	0	4	0	0	0	2	0	6	0
47.5	1	0	0	0	2	0	3	0	4	0	1	0	3	0	8	0
48.5	3	0	0	0	2	1	5	1	0	0	3	0	0	0	3	0
49.5	2	0	0	0	1	0	3	0	0	0	0	0	0	0	0	0
50.5	2	0	0	0	3	0	5	0	4	0	2	0	2	0	8	0
51.5	0	0	0	0	2	0	2	0	2	0	1	0	1	0	4	0
52.5	2	0	0	0	0	0	2	0	2	0	0	0	1	0	3	0
53.5	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0
54.5	1	0	0	0	0	0	1	0	0	0	0	0	2	0	2	0
55.5+	1	0	0	0	3	0	4	0	5	0	4	0	4	0	13	0

Table 7. Selection curve parameter estimates and hypothesis tests obtained from the covered cod-end data of Table 6, fitted to the total catch data for 20 min. and 60 min. duration hauls. The logistic curve corresponds to $\delta = 1$. Model fit was assessed from Figure 5 and using deviance and d.f.'s calculated using only those length classes with expected catch of greater than 3 in both cod end and cover. Deviance and d.f.'s calculated using all data are shown in square brackets, and are used to test $H_0 : \delta = 1$. Standard errors (based on binomial theory) are given in parentheses for the acceptable fits. Standard errors for the 60 min. hauls must be corrected for between-haul variation (see text, 'Combined hauls analysis')

	20 min. hauls, all lengths		60 min. hauls, all lengths		60 min. hauls, subset of lengths	
	Logistic curve	Richards curve	Logistic curve	Richards curve	Logistic curve	Richards curve
a	−10.6 (0.9)	−14.5	−9.0	−25.7 (7.0)	−13.7 (1.1)	−14.9
b	0.30 (0.02)	0.39	0.25	0.62 (0.16)	0.37 (0.03)	0.39
δ		1.6		3.7 (1.1)		1.2
l_{25}	31.4 (0.6)	31.8	32.0	33.2 (0.3)	34.2 (0.4)	34.2
l_{50}	35.0 (0.4)	35.5	36.5	37.4 (0.3)	37.2 (0.2)	37.2
l_{75}	38.6 (0.5)	38.7	41.0	40.4 (0.3)	40.1 (0.3)	40.1
H_0 : model						
Deviance	12.7 [23.4]	10.8 [22.3]	36.5 [46.9]	17.9 [25.4]	8.3 [10.6]	8.2 [10.5]
d.f.	12 [34]	11 [33]	20 [33]	17 [32]	11 [25]	11 [24]
P value	0.39	0.46	0.01	0.40	0.69	0.69
$H_0 : \delta = 1$						
Deviance		1.1		21.5		0.1
d.f.		1		1		1
P value		0.3		< 0.001		0.99

on the objectives of the study. The Richards fit to the full data is the best model of the entire selection curve and is effective at quantifying the capture of the smaller fish. The logistic fit to the reduced data might be acceptable if only l_{50} and *SR* are of interest. These two fits give similar inferences about l_{50} and *SR*; in particular, the estimated l_{50}'s differ by only 0.2 cm.

Variation between replicate deployments

A selectivity experiment usually requires repeated deployment of the gear, ideally over a range of conditions similar to those anticipated in typical commercial operation. Particularly in towed gear selectivity experiments, there are sufficient catch numbers for a selection curve to be estimated for each replicate deployment of the gear. Such analysis of data from replicate hauls has led to the widely accepted view that, in practice, the selectivity of a cod end changes from haul to haul even though the gear has not been altered (Fryer, 1991). We are not aware of any information about variation in selectivity between deployments of traps, gill nets or hooks. This information may be difficult to obtain because catch numbers are generally low (relative to towed gears) and because nonselective variants can generally not be used.

The causes of between-haul variation are unclear, but could arise from changes in uncontrolled variables such as towing direction, wind speed, water depth, or spatial structure of stock. Although between-haul variation is often not of direct interest, it must be accounted for in the analysis of selectivity data to avoid making misleading statistical inferences about the selectivity parameters. Fryer (1991) demonstrates some of the inferential problems incurred by ignoring between-haul variation.

Various approaches for dealing with between-haul variation are described below. Because the problem is most pertinent for towed gear studies, it is convenient to consider the specific example of a covered cod-end experiment with H deployments ('hauls'). Only minor changes are required for trouser trawl or other gear experiments.

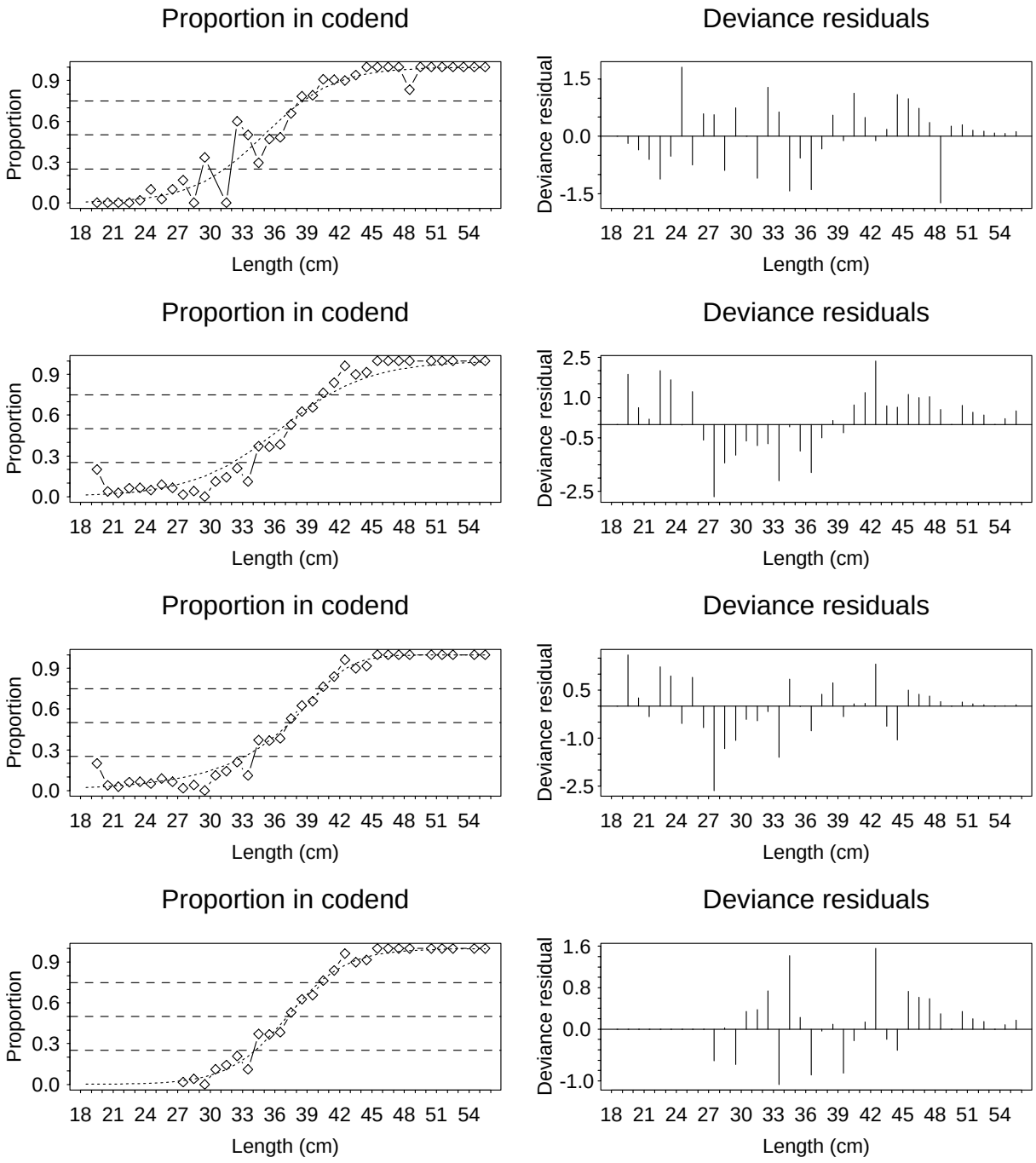


Figure 5. Fits and residuals for the covered cod-end data (totals) of Table 6. From top to bottom, logistic curve fit to 20. min haul data; logistic curve fit to 60. min haul data; Richards curve fit to 60. min haul data; logistic curve fit to subset of 60 min. haul data.

Combined hauls analysis

One approach is to estimate the selection parameters as if the catch data had come from a single haul ('Covered cod-end haddock data', above). For haul h, let n_{l1}^h and n_{l+}^h be the number of length l fish in the cod end and cod end plus cover, respectively. A combined hauls analysis uses the proportion of length l fish taken by the cod end over the H hauls,

$$y_{l\cdot} = \frac{\sum_h n_{l1}^h}{\sum_h n_{l+}^h}, \tag{15}$$

and can be viewed as estimating the 'mean' selection curve of the experimental gear in the sense that it estimates the selectivity of the experimental gear with respect to the population of fish it will contact under repeated deployment. However, inferences must allow for any between-haul variation, and two methods (REP and bootstrapping) for doing so are given below. A limitation of these methods is that they are not suitable for model selection, but rather, for correcting standard errors and confidence intervals concerning the estimated parameters of the 'mean' selection curve.

Replication estimate of dispersion (REP)

The replication estimate of dispersion (*REP*) is defined to be (McCullagh and Nelder, 1989)

$$REP = \frac{Q}{d.f.}, \tag{16}$$

where:

$$Q = \sum_l \sum_h \frac{(n_{l1}^h - n_{l+}^h y_{l\cdot})^2}{n_{l+}^h y_{l\cdot}(1 - y_{l\cdot})}, \tag{17}$$

where d.f. $= (H - 1) \times$ the number of length classes in the summation. (In practice, the summation would be restricted to length classes for which $\sum_h n_{l+}^h$ is reasonably large and $y_{l\cdot}$ is not too close to 0 or 1.) Under the null hypothesis of no between-haul variation, Q has an approximate chi-square distribution on d.f. degrees of freedom. If this null hypothesis is rejected, standard errors of estimated parameters should be multiplied by $\sqrt{REP}$.

For example, the individual-haul haddock data from Clark (1957) (Table 6) can be used to determine if the standard errors from the combined haul analysis (Table 7) should be adjusted. Using only length classes with $0.1 \leq y_{l\cdot} \leq 0.9$, $Q = 15.9$ on 18 d.f. for the 20 min. hauls, and no between-haul variation is indicated. However, $Q = 48.1$ on 27 d.f for the 60 min. hauls, corresponding to $REP \approx 1.78$. This is significant (P value < 0.01) and the standard errors of the estimated parameters should be multiplied by $\sqrt{1.78} \approx 1.33$.

Bootstrapping

Bootstrapping (Efron and Tibshirani, 1993) is a computer-intensive Monte-Carlo technique that aims to emulate the variability (including between-haul variability) in selectivity data. Bootstrapping should work well provided there are a reasonable number of hauls; the three hauls in Clarks' data are probably insufficient. Millar (1993b) used the bootstrap in a combined haul analysis of data from a scallop dredge.

Subsampling

When catches have been subsampled, it is necessary to estimate the total catch at length for each haul (using the subsampling fractions) before combining the data over hauls. This ensures that large catches contribute more to the mean selection curve than small catches. *REP* can then be calculated as before, but it now represents extra variation owing to both between-haul variation and subsampling. When there are a reasonable number of hauls, bootstrapping should again work well, because it can naturally incorporate the additional variability introduced by subsampling.

Fixed and random effects model

An alternative to the combined hauls approach is to model the between-haul variation explicitly through analysis of individual-haul data. If the true selection curve of each haul can be described by a q parameter model, then denote the selection parameters for haul h by $\nu_h = (\nu_{h1}, \ldots, \nu_{hq})^T$. For example, for a logistic curve, $q = 2$ and $\nu_h = (a_h, b_h)^T$; similarly, for a Richards curve, $\nu_h = (a_h, b_h, \delta_h)^T$. In some cases it may be enough to work with just a single selectivity parameter from each haul, $l_{50,h}$, say.

The mixed (i.e. fixed and random) effects selectivity model of Fryer (1991) accounts for between-haul variation by treating it as a random effect, in which selection varies randomly between hauls about a mean selection curve with a certain probability distribution. Controlled changes to the gear (e.g. changes in mesh size) are also modelled, as fixed effects, by allowing the mean selection curve to change in a systematic way. It is assumed that

the selectivity parameters vary from haul to haul according to the model

$$\nu_h = \theta_h + \epsilon_h \tag{18}$$

where:

- θ_h is a $q \times 1$ vector of unknown parameters giving the expected selection parameters for the gear used in haul h. Fryer (1991) and Wileman et al. (1996) show how θ_h is used to incorporate fixed effects (changes in construction or deployment of the trawl) into the model. If the hauls are true replicates (no fixed changes in construction or deployment of the gear) then θ_h can be replaced by θ for all hauls.
- ϵ_h is a $q \times 1$ error vector, representing the random variation in selectivity for haul h.

The ϵ_h are assumed to be independent and multivariate normally distributed with zero mean and constant, but unknown, $q \times q$ variance matrix D. Thus, ν_h has a multivariate normal distribution with mean θ_h and variance D; i.e.

$$\nu_h \sim \mathrm{N}_q(\theta_h, \mathrm{D}). \tag{19}$$

Note that D = 0 corresponds to a fixed-effects-only model.

The mixed effects model is a special case of the Laird–Ware model for longitudinal data (Laird and Ware, 1982; see also Jones, 1993). Other types of mixed models are feasible, particularly ones where the variance matrix changes with the controlled changes to the net, but these are not considered here.

Parameter estimation

In principle, a likelihood function for θ_h and D can be derived and estimates obtained by maximizing this likelihood (Stiratelli et al., 1984). However, this approach is complicated and computationally intensive. (At the time of writing, built-in facilities for the fitting of generalized mixed models are becoming available in some statistical software packages and these should allow fitting of the mixed effects model in simple cases such as a logistic selection curve to covered cod-end data.) An alternative, which is usually adequate, is to use the selection parameters estimated from each individual haul, and this is described below (Korn and Whittimore, 1979; Laird and Ware, 1982; Stiratelli et al., 1984).

Let $\hat{\nu}_h$ be the maximum likelihood estimator of ν_h and let R_h be its estimated $q \times q$ variance matrix. (Note that any subsampling is automatically incorporated in the estimation of $\hat{\nu}_h$ and R_h; 'General model', above) Assume that a sufficient number of fish at length were measured so that $\hat{\nu}_h$ is approximately multivariate normally distributed with mean ν_h and variance matrix R_h. That is, conditional on ν_h,

$$\hat{\nu}_h \sim \mathrm{N}_q(\nu_h, \mathrm{R}_h). \tag{20}$$

Combining this with Expression 19 includes the randomness in ν_h, giving

$$\hat{\nu}_h \sim \mathrm{N}_q(\theta_h, \mathrm{R}_h + \mathrm{D}). \tag{21}$$

In Expression 21, the variance matrix D measures the between-haul variation in the selectivity parameters ν_h and the variance matrices R_h measure the 'known' variation in estimating $\hat{\nu}_h$ owing to the randomness in the proportions of fish observed. Estimates of θ_h and D can then be obtained by either maximum likelihood or residual maximum likelihood (Robinson, 1987). Both sets of estimates can be obtained either directly using a numerical maximization routine or by the EM algorithm (Dempster et al., 1977). For details, see Fryer (1991).

Applications of the mixed model in the selectivity literature are Reeves et al. (1992) and Galbraith et al. (1994).

Fixed-effects-only approach

A number of studies of individual-haul selectivity have attempted to explain the between-haul variation using measurements (e.g. catch size) taken from the haul during deployment and regressing these against the individual-haul parameters (Hodder and May, 1964; Suuronen et al., 1991; Erickson et al., 1996; Neill and Kynoch, 1996). These studies are not in universal agreement, but the majority support the view that selectivity decreases (i.e. l_{50} increases) with increasing catch size. However, there typically remains substantial unexplained between-haul variation which must be accounted for before making further inference.

Covered cod-end example, revisited

Logistic selection curves were fitted (Table 8) to the individual-haul data (Table 6). The fits to the 60 min. data were restricted to lengths above 27 cm (as in 'Covered cod-end haddock data', above).

Although the gear used does not vary, the difference between 20 min. and 60 min. hauls can be considered a fixed effect under the control of the experimenter.

Table 8. Individual haul fits to the data of Table 6

Haul	Duration (min)	$\hat{a}$ (SE)	$\hat{b}$ (SE)	corr($\hat{a}, \hat{b}$)	$\hat{l}_{50}$ (SE)	$\hat{SR}$ (SE)
1	20	−26.4 (7.1)	0.735 (0.191)	−0.9988	35.9 (0.58)	2.99 (0.78)
2	20	−11.5 (2.7)	0.325 (0.071)	−0.9934	35.6 (1.01)	6.77 (1.49)
3	20	−9.82 (0.9)	0.279 (0.025)	−0.9853	35.3 (0.55)	7.89 (0.71)
4	60	−11.6 (1.3)	0.307 (0.035)	−0.9958	37.6 (0.40)	7.15 (0.82)
5	60	−15.3 (2.9)	0.427 (0.076)	−0.9964	35.7 (0.63)	5.14 (0.92)
6	60	−17.8 (2.5)	0.476 (0.066)	−0.9979	37.4 (0.36)	4.62 (0.64)

Mixed model results

Here, we model the individual-haul parameter estimates $\hat{v}_h = (\hat{a}_h, \hat{b}_h)$ of Table 8 using model (21). We assume that the mean selection curves for 20 min. and 60 min. hauls are logistic with parameters $\theta = (a^{(20)}, b^{(20)})$ and $\theta = (a^{(60)}, b^{(60)})$, respectively. Four models were considered.

1. $a^{(20)} = a^{(60)}, b^{(20)} = b^{(60)}$: same mean selection curve for both 20 and 60 min tows.
2. $a^{(20)} = a^{(60)}$: same relative change in l_{50} and selection range.
3. $b^{(20)} = b^{(60)}$: common selection range, but different l_{50}'s.
4. no constraints on $a^{(20)}, b^{(20)}, a^{(60)}, b^{(60)}$: different mean selection curves.

Inspection of the log-likelihoods from the maximum likelihood fits (Table 9) shows that models 2 and 3 are preferred. Model 2 and model 3 have the same number of parameters, so model 3 is chosen as best because of its slightly higher log-likelihood. Thus, haul duration has no effect on selection range, but does on 50% retention length. Similar results were obtained from the restricted maximum likelihood method.

Model 3 estimates a common mean *SR* of 6.1 cm and mean l_{50}'s of 35.1 and 37.1 cm for the 20 min. and 60 min. hauls respectively. The corresponding standard errors are 0.6, 0.4, and 0.4 cm, respectively. The between-haul variation is estimated to be

$$\mathrm{D} = \begin{pmatrix} 4.678 & -0.1365 \\ -0.1365 & 0.003986 \end{pmatrix}. \qquad (22)$$

Fixed effects model results

A one-way ANOVA of haul duration (explanatory factor) on $\hat{l}_{50}$'s (response) from Table 8 estimated the mean l_{50}'s of 20 min. and 60 min. tows to be 35.6 cm and 37.3 cm respectively. The ANOVA was weighted using the inverse of the estimated variance of each $\hat{l}_{50}$. With these weights, the residual variance estimated from the ANOVA should be close to unity. In fact, it was 1.94, providing modest evidence (*P* value = 0.1) of between-haul variation. Using the residual variance of 1.94 implicitly incorporates some component of between-haul variation albeit in an *ad-hoc* way, and gives a conservative analysis, with the estimated l_{50}'s having standard errors of 0.5 and 0.3, respectively, and haul duration having a marginally significant (*P* value = 0.05) effect on l_{50}. Taking a residual variance of unity would ignore any between-haul variation, would reduce the standard errors, and would make the effect of haul duration 'more significant'. A similar analysis found no effect of haul duration on selection range (*P* value > 0.1).

Design of (towed-gear) experiments

Good experimental design is necessary to ensure that the objectives of the experiment are likely to be met. Good design also ensures value for money: the experiment is not a waste of resources because nothing can be gleaned from it, nor is it profligate by collecting unnecessary data. In selectivity studies, good design involves ensuring an adequate number of deployments, and an adequate number of fish measured, to meet the aims of the experiment. Typical aims might be to estimate a selection curve with sufficient precision, or to detect any important differences between two gears, should they occur. Clearly, what is meant by 'sufficient' precision and an 'important' difference will depend on the requirements of management.

The theory of experimental design is a huge topic, and depends on the experiment's objectives, the types of variation in the data, the statistical model to be

Table 9. Fit of the four mixed effect models described in the text

Model	d.f.	Log-likelihood	$\hat{a}^{(20)}$	$\hat{b}^{(20)}$	$\hat{a}^{(60)}$	$\hat{b}^{(60)}$	$\hat{l}_{50}^{(20)}$	$\hat{SR}^{(20)}$	$\hat{l}_{50}^{(60)}$	$\hat{SR}^{(60)}$
1	7	1.85	−12.7	0.352	−12.7	0.352	36.1	6.2	36.1	6.2
2	6	4.64	−13.2	0.374	−13.2	0.355	35.3	5.9	37.2	6.2
3	6	4.89	−12.6	0.360	−13.3	0.360	35.0	6.1	36.9	6.1
4	5	5.41	−11.0	0.314	−14.0	0.381	35.0	7.0	36.7	5.8

used, and so on. Here, we consider just one example, based on the mixed model used above ('Fixed and random effects model'); namely that of estimating the mean selection curve θ of an experimental cod end by making H hauls. In particular, we investigate the effect of changing both the number of hauls H, and the level of subsampling from each haul. The theory can easily be placed in the context of other gear experiments.

To proceed, we assume that the selection parameters for each haul, v_h, are estimated with constant variance R. This is not too unreasonable if, after each haul, there is a fixed time allocated for measuring fish and there are always plenty of fish. Thus, the catches are subsampled, with approximately the same number of fish measured each haul. It turns out that the assumption isn't critical, but it does make the theory tractable. Using the model mentioned above,

$$\hat{v}_h \sim N_q(\theta, \mathrm{D} + \mathrm{R}) \tag{23}$$

and θ can be estimated by:

$$\hat{\theta} = \frac{1}{H}\sum_h \hat{v}_h \tag{24}$$

which has distribution

$$\hat{\theta} \sim N_q\left(\theta, \frac{\mathrm{D} + \mathrm{R}}{H}\right). \tag{25}$$

Thus, to estimate the selection parameters θ sufficiently precisely, the variance matrix $(\mathrm{D} + \mathrm{R})/H$ must be made suitably 'small', most obviously by choosing an appropriate number of hauls H and measuring enough fish each haul.

In practice, D and R are unknown and must be estimated from the data, so inferences about $\hat{\theta}$ should be based on Hotelling's T^2-distribution (the multivariate generalization of Student's t-distribution), or its F-distribution equivalent, rather than the multivariate normal distribution. The usual estimate of D + R is the $q \times q$ matrix S given by:

$$\mathrm{S} = \frac{1}{H-1}\sum_h (\hat{v}_h - \hat{\theta})(\hat{v}_h - \hat{\theta})^T \tag{26}$$

and then

$$\frac{H(H-q)}{q(H-1)}(\hat{\theta} - \theta)^T \mathrm{S}^{-1}(\hat{\theta} - \theta) \sim F(q, H-q), \tag{27}$$

where $F(q, H-q)$ is an F-distribution on q and $H-q$ degrees of freedom (Chatfield and Collins, 1980).

To use this to design a selectivity study, *a priori* knowledge is needed both of the appropriate form of the selection curve, and of the variances D and R. These, of course, won't be known until after the experiment. The usual practice is either to conduct a pilot study (before a large experiment) or to use estimates from previous studies.

For example, consider investigating a cod end similar to that in Clark's 1957 experiment. Assuming the selection curve to be logistic, so that $\theta = (a, b)^T$, we could then use variance estimates from Clark's study. That is, we could take the between-haul variance to be D given in Equation 22, and the within-haul variance to be, say, the 'median' variance for the 60 min. hauls (given by haul 6):

$$\mathrm{R} = \begin{pmatrix} 6.33 & -0.165 \\ -0.165 & 0.00434 \end{pmatrix}. \tag{28}$$

We also need 'ballpark' estimates of the selection parameters, and sensible values here are $(-13.3, 0.359)^T$ (from Table 9). These estimates can then be plugged into Expression 27 to show how confidence regions for θ depend on the number of hauls. It is then simple to transform these regions into confidence regions for $(l_{50}, SR)^T$. For example, ten hauls would estimate l_{50} to within about 1.5 cm, and SR to within about 3 cm with 95% confidence (Figure 6).

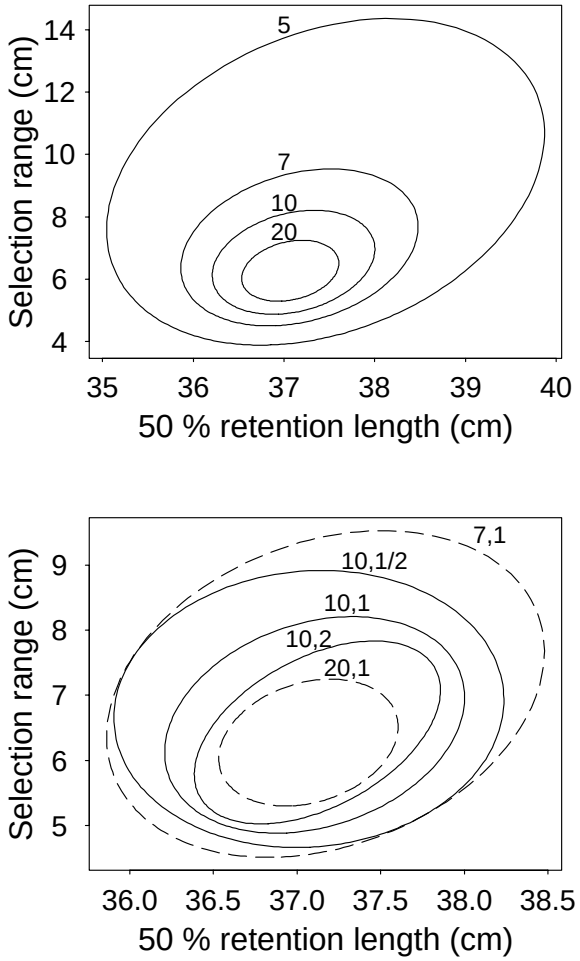


Figure 6. Top plot shows the 95% confidence regions for $(l_{50}, SR)^T$ for H = 5, 7, 10, and 20 hauls. The number next to each region indicates the value of H. Bottom plot shows the confidence regions for $(l_{50}, SR)^T$ when H = 10 hauls, for status quo effort (k = 1), and when sampling effort is halved (k = 1/2) and doubled (k = 2). The dashed lines show similar confidence regions when H = 7, 20, for status quo effort. The pair of numbers next to each region indicates the value of H,k.

The above example assumes it will be possible to measure a similar number of fish as caught in Clark's 60 min. hauls. This could be considered as 'status quo' sampling effort. The effect of measuring more fish (if catches permit) can be assessed by noting that increasing the sampling effort by a factor of k, say, reduces the within-haul variance to R/k. Similarly, measuring fewer fish (changing sampling effort by a factor of $1/k$), increases the within-haul variance to kR. Here, halving the number of fish measured per haul enlarges the confidence region for $(l_{50}, SR)^T$ substantially (Figure 6). Measuring twice as many fish per haul would have some benefit, but if time is limiting between hauls, then it would be more efficient to have 20 hauls with status quo effort, rather than 10 hauls with double the sampling effort (although this would be environmentally less sound).

The above example is only intended for illustration. In particular, the variances are estimated on few degrees of freedom, so the results should not be over interpreted. However, in our experience, the qualitative conclusions appear true for towed gear experiments in general. Ensuring an adequate number of hauls is critical; measuring more fish on each haul has a comparatively small effect.

Conclusion

The statistical analysis of comparative size-selectivity experiments is now on a firm foundation. This required precise specification of 'size-selectivity' and the use of appropriate statistical techniques. In addition to answering some questions (e.g. how to incorporate different fishing efficiencies of gear variants; subsampling; between-haul variation), the clear conceptualization of the size-selectivity process has better defined the limitations of this methodology and identified where further research is required. In particular, areas worthy of further work include:

- better understanding of fish behaviour, and hence of the relationship between population-selectivity and contact selectivity;
- appropriate methodology for analysis of pseudo-direct studies (these correspond to indirect studies in which the population length distribution is estimated independently of the selectivity experiment);
- model 'identifiability' for gill net and hooks, i.e. the phenomenom that two quite distinct selectivity models may be totally indistinguishable because they produce the same fitted values η_{jl} (Millar, 1995);
- applicability to commercial trawling conditions (e.g. effect of catch size: Cook and Reeves, 1996; Erickson et al., 1996).

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